

Differential Equation

Definition: The equation which involve one dependent variable and its derivative with respect to one or more independent variable is called a differential equation.

Example: $\frac{d^2y}{dx^2} + p^2y = 0$

$$\frac{\partial^2 y}{\partial t^2} = a^2 \frac{\partial^2 y}{\partial x^2}$$

Classification: There are two types of differential equation: They are - 1) Ordinary diff. eqn

2) Partial diff. eqn

Ordinary Differential Eqn: The differential eqn which involve only one independent variable is called ordinary differential eqn. Ex: 1) $\frac{d^2y}{dx^2} + p^2y = 0$

$$2) \frac{dy}{dx} = \frac{\sqrt{1-x^2}}{\sqrt{2-y^2}}$$

Partial Differential Eqn: The differential eqn which involve partial differential co-efficient w.r. to more than one independent variable

is called partial differential eqn.

Ex: 1) $\frac{\partial^2 y}{\partial t^2} = a^2 \frac{\partial^2 y}{\partial x^2}$

2) $\frac{\partial z}{\partial x} + \frac{\partial x}{\partial y} = kz$

Linear and non Linear Differential eqn:

A differential eqn in which the dependent variable and all its derivatives present occur in which the first degree only and no product of dependent variables and/or derivatives occur is known as a linear differential equation.

A differential equation which is not linear is called a non linear differential eqn.

2 condition (any one) for standard formation of differential equation

1. যে ফ^n তে আমরা থাকলে যদি differentiation করতে করতে যদি Given ফ^n এর arbitrary constant remove হয়ে যায়।

Ex: $y = A \cos x (px - \alpha)$; here A is an arbitrary constant

$$\therefore \frac{dy}{dx} = -A \sin x$$

$$\therefore \frac{d^2y}{dx^2} = -A \cos x = -y$$

$$0\pi, \frac{d^2y}{dx^2} + y = 0$$

২। যে f^n দেওয়া থাকবে সেভাবে করতে করতে যদি এটি বানিয়ে দিতে পারি.

Q: Form the differential eqn whose solution is given/eliminate the constant
 $y = A \cos (px - \alpha)$

Soln: Given that,

$$y = A \cos (px - \alpha) \quad \text{————— (1)}$$

Differentiating the given eqn we have,

$$\frac{dy}{dx} = -AP \sin (px - \alpha)$$

Again,

$$\begin{aligned} \frac{d^2y}{dx^2} &= -AP^2 \cos (px - \alpha) \\ &= -P^2 y \end{aligned}$$

$\therefore \frac{d^2y}{dx^2} + P^2 y = 0$; which is the required eqn.

Q: Form the differential eqn whose solution is $y = A \cos x + B \sin x$

Solⁿ: Given that,

$$y = A \cos x + B \sin x \quad \text{--- (1)}$$

$$\therefore \frac{dy}{dx} = -A \sin x + B \cos x$$

$$\begin{aligned} \text{Again, } \frac{d^2y}{dx^2} &= -A \cos x - B \sin x \\ &= -(A \cos x + B \sin x) \\ &= -y \quad [\text{By eqn (1)}] \end{aligned}$$

$$\text{or, } \frac{d^2y}{dx^2} + y = 0 \text{ which is the required eqn.}$$

Here $\frac{d^2y}{dx^2} + y = 0$ is a eqn and $y = A \cos x + B \sin x$ is the solⁿ of the eqn.

Ex-1 Eliminate the constant from $y = ax + bx^2$

Solⁿ: Given that,

$$y = ax + bx^2$$

$$\therefore \frac{dy}{dx} = a + 2bx \quad \text{--- (1)}$$

$$\therefore \frac{d^2y}{dx^2} = 2b \quad \text{--- (2)}$$

From eqn (1),

$$a = \frac{dy}{dx} - 2bx = \frac{dy}{dx} - x \frac{d^2y}{dx^2} \quad [\text{By eqn (2)}]$$

Putting the value of a and b in the given eqn we get,

$$y = \left(\frac{dy}{dx} - x \frac{d^2y}{dx^2} \right) x + b \left(\frac{1}{2} \cdot \frac{d^2y}{dx^2} \right) x^2$$

$$\text{or, } 2y = 2x \frac{dy}{dx} - 2x^2 \frac{d^2y}{dx^2} + x^2 \frac{d^2y}{dx^2}$$

$$\text{or, } x^2 \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} + 2y = 0 ; \text{ which is the}$$

required eqn.

Ex-3: Form the differential eqn of which $c(y+c)^2 = x^3$ is the complete integral.

Soln: Given, $c(y+c)^2 = x^3 \quad \text{--- (1)}$

Differentiating eqn (1) we have,

$$2c(y+c) \cdot \frac{dy}{dx} = 3x^2 \quad \text{--- (2)}$$

Dividing (1) by (2) we get, $\frac{y+c}{2 \frac{dy}{dx}} = \frac{x}{3}$

$$\text{On, } y+c = \frac{2x}{3} \cdot \frac{dy}{dx}$$

$$\text{On, } c = \frac{2x}{3} \frac{dy}{dx} - y$$

Putting the value of c in eqn (2) we get,

$$2 \left(\frac{2x}{3} \cdot \frac{dy}{dx} - y \right) \cdot \left(\frac{2x}{3} \frac{dy}{dx} \right) \cdot \frac{dy}{dx} = 3x^2$$

$$\text{On, } \left(12x \frac{dy}{dx} - 18y \right) \left(\frac{2x}{3} \left(\frac{dy}{dx} \right)^2 \right) = 27x^2$$

$$\text{On, } 8x^2 \left(\frac{dy}{dx} \right)^3 - 12xy \left(\frac{dy}{dx} \right)^2 = 27x^2$$

$$\therefore 8x \left(\frac{dy}{dx} \right)^3 - 12y \left(\frac{dy}{dx} \right)^2 = 27x$$

which is the required equation.

Q: Form the differential equation corresponding to the family of curves $y=c(x-c)^2$.

Soln: Given that,

$$y=c(x-c)^2 \quad \text{--- (1)}$$

Differentiating eqn (1) we get,

$$\frac{dy}{dx} = 2c(x-c) \quad \text{--- (2)}$$

From eqn (1) and (2) we get,

$$\frac{y}{\frac{dy}{dx}} = \frac{x-c}{2}$$

$$\text{or, } x-c = \frac{2y}{\frac{dy}{dx}}$$

$$\therefore c = x - \frac{2y}{\frac{dy}{dx}}$$

Putting the value of c in eqn (2) we get,

$$\frac{dy}{dx} = 2 \left(x - \frac{2y}{\frac{dy}{dx}} \right) \left(\frac{2y}{\frac{dy}{dx}} \right)$$

$$\text{or, } \left(\frac{dy}{dx} \right)^3 = 2 \left(x \cdot \frac{dy}{dx} - 2y \right) \cdot 2y = 4y \left(x \cdot \frac{dy}{dx} - 2y \right)$$

which is the required differential equation.

Ex-5 Find the differential eqn of all circles passing through the origin and having their centres on the x -axis.

Soln: Equation of circle passing through the origin and having their centres on the x -axis is,

$$x^2 + y^2 + 2gx = 0 \quad \text{————— (1)}$$

differentiating eqn (1) we get,

$$2x + 2y \frac{dy}{dx} + 2q = 0$$

$$\text{or, } x + y \frac{dy}{dx} + q = 0$$

$$\therefore q = -\left(x + y \frac{dy}{dx}\right)$$

putting these value q is eqn (1) we get,

$$x^2 + y^2 + 2 \cdot \left(-\left(x + y \frac{dy}{dx}\right)\right) x = 0$$

$$\text{or, } y^2 = x^2 + 2xy \frac{dy}{dx}$$

which is the required differential eqn.

* Eqn এর order বরা বরা আগে তাকে linear form এ নিতে হবে

* Order হলো অর্ধেক বরাবর diff. করাছি

* Degree " " মতাবর diff. মতো মাঝে মাঝে তার উল্লেখ হয় নাওমাঝে অনিই ,

* Degree rational number (অমূলক আকারে)

থাকতে পারে, হবে,

Order: The order of a differential equation is the order of the highest derivative involved.
Order is defined if we can express the given eqn as a linear equation.

Degree: The degree of a differential equation is the power of the highest derivative after the equation has been made rational.

Ex 10 (11) Form the differential eqn if $y = \cancel{ae^{2x} + be^{-3x}} + c \cos x$
 $y = a \cos(mx + b)$

Soln: Given that,

$$y = a \cos(mx + b) \quad \text{--- (1)}$$

$$\therefore \frac{dy}{dx} = -a m \sin(mx + b)$$

$$\therefore \frac{d^2y}{dx^2} = -a m^2 \cos(mx + b) = -m^2 y \quad [\text{By eqn (1)}]$$

$$\text{or, } \frac{d^2y}{dx^2} + m^2 y = 0$$

which is the required eqn.

$$y = Aex + Be^{-x} + x^2.$$

Soln : Given that,

$$xy = Aex + Be^{-x} + x^2 \quad \text{--- (1)}$$

$$\therefore y + x \frac{dy}{dx} = Aex - Be^{-x} + 2x$$

$$\therefore \frac{dy}{dx} + \frac{dy}{dx} + x \frac{d^2y}{dx^2} = Aex + Be^{-x} + 2$$

$$\text{Or, } Aex + Be^{-x} = 2 \frac{dy}{dx} + x \frac{d^2y}{dx^2} - 2 \quad \text{--- (2)}$$

From eqn (1) and (2) we get,

$$xy = 2 \frac{dy}{dx} + x \frac{d^2y}{dx^2} - 2 + x^2$$

$$\text{Or, } x \frac{d^2y}{dx^2} + 2 \frac{dy}{dx} = xy - x^2 + 2$$

which is the required differential eqn.

Q: Find the order and degree of the following diff. eqn.

$$\textcircled{1} \sin\left(\frac{dy}{dx}\right) + x = 0$$

Soln: Given that,

$$\sin \sin \left(\frac{dy}{dx} \right) + x = 0$$

$$\text{or, } \frac{dy}{dx} = \sin^{-1}(\sin - x)$$

which is linear and rational.

order = 1 and degree = 1.

$$11) \frac{dy}{dx} + \sin \frac{dy}{dx} + x + 2y = 0$$

Soln: Given that,

$$\frac{dy}{dx} + \sin \frac{dy}{dx} + x + 2y = 0$$

The equation can not be expressed as linear eqn. so order and degree of the eqn is not defined.

* $y = A \cos x + B \sin x$ is called the solution of

$$\frac{d^2y}{dx^2} + y = 0$$

General solution: সমাধান এর মধ্যে অর্ধেক constant থাকবে.

$$\text{Ex: } y = A \cos x + B \sin x$$

The relation containing n arbitrary constant which satisfies an ordinary differential equation of n th order is called its complete or general solution.

Particular solution: A solution of an ordinary differential equation without arbitrary constant.

Differential eqn of First order and First

Degree

$$M + N \frac{dy}{dx} = 0$$

M and N are fn of x and y or constant.

Method of Solution: 1) variable separable

2) Homogeneous Differential equation

3) Linear Differential Equation

4) Exact Differential Equation

variable Separable Method:

Working rules: ২টা variable চাওয়া থাকবে যেমন x, y . variable ২টা চক্ক দুইদিকে নিয়ে আলাদা করবে

$$\text{If } M(x, y)dx + N(x, y)dy = 0$$

$$\text{Let, } M(x, y) = f(x) \cdot g(y)$$

$$\& N(x, y) = \pi(x) \cdot S(y)$$

so given eqn becomes, $f(x) \cdot g(y) \cdot dx + \pi(x) \cdot S(y)dy = 0$

$$\therefore \frac{f(x)}{\pi(x)} dx + \frac{S(y)}{g(y)} dy = 0$$

Now integrating we will get our required solution.

Ex: 2 | Solve $\frac{dy}{dx} = e^{x-y} + x^2 e^{-y}$

Solution: Given that,

$$\frac{dy}{dx} = e^{x-y} + x^2 e^{-y}$$

$$\text{or, } \frac{dy}{dx} = \frac{e^x}{e^y} + \frac{x^2}{e^y} = \frac{e^x + x^2}{e^y}$$

$$\text{or, } e^y dy = (e^x + x^2) dx$$

Now integrating, $\int e^y dy = \int (e^x + x^2) dx$

or, $e^y + c = e^x + \frac{x^3}{3}$; which is the required solution.

এম প্রাক্ষর মন কাম মে প্রাক্ষর a constant টে আলাদা
তয় কাম, তা ব্যবলও সমস্যা লেই.

Ex: 1 Solve $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$

Soln: Given that,

$$\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$$

$$\text{or, } \frac{dy}{1+y^2} = \frac{dx}{1+x^2}$$

Now integrating, $\int \frac{dy}{1+y^2} = \int \frac{dx}{1+x^2}$

$$\text{or, } \tan^{-1} y = \tan^{-1} x + c$$

which is the required solution.

Ex-3 Solve $\sec^2 x \tan y dx + \sec^2 y \tan x dy = 0$

Soln: Given,

$$\sec^2 x \tan y dx + \sec^2 y \tan x dy = 0$$

$$\text{or, } \frac{\sec^2 x \tan y dx + \sec^2 y \tan x dy}{\tan x \tan y} = 0$$

$$\text{or, } \frac{\sec^2 x}{\tan x} dx + \frac{\sec^2 y}{\tan y} dy = 0$$

Now Integrating, $\log \tan x + \log \tan y = 0 + C$

which is the required solution.

Ex-4 | Solve $(y - px)x = y$ here $p = \frac{dy}{dx}$

Soln: Given that,

$$(y - px)x = y$$

$$\text{or, } xy - x^2 p = y$$

$$\text{or, } x^2 p = xy - y = y(x - 1)$$

$$\text{or, } \frac{x^2}{x-1} p = y$$

$$\text{or, } \frac{x^2}{x-1} \cdot \frac{dy}{dx} = y$$

$$\text{or, } \frac{dy}{y} = \frac{x-1}{x^2} dx = \left(\frac{1}{x} - \frac{1}{x^2} \right) dx$$

Now Integrating, $\ln y + C = \ln x + \frac{1}{x}$

which is the required solution.

Ex-5 | Solve $y - x \frac{dy}{dx} = a(y^2 + \frac{dy}{dx})$

Soln: Given that, $y - x \frac{dy}{dx} = a(y^2 + \frac{dy}{dx})$

$$\text{or, } y - ay^2 = a \frac{dy}{dx} + x \frac{dy}{dx}$$

$$y(1-ay) = (a+x) \frac{dy}{dx}$$

$$\text{or, } \frac{dx}{x+a} = \frac{dy}{y(1-ay)}$$

$$= \left(\frac{1}{y} + \frac{a}{1-ay} \right) dy$$

$$= \left(\frac{1}{y} - \frac{-a}{1-ay} \right) dy$$

Now Integrating,

$$\ln(x+a) = \ln y - \ln(1-ay) + \ln c$$

$$= \ln \frac{y}{1-ay} \cdot c$$

$$\text{or, } x+a = e^{\ln \left(\frac{y}{1-ay} \right) \cdot c}$$

$$\therefore x+a = c \cdot \frac{y}{1-ay}$$

which is the required solution.

Equation reducible to variable separable method

যদি অজ্ঞেয় অর্ধক $ax+by+c$ আকারে
 বসানো অবিচ্ছিন্ন থাকবে তখন এই method

use করা,

$$\frac{1}{y} + \frac{1}{(1-ay)}$$

$$y=0 \rightarrow (1-a \cdot 0) = 1$$

$$1-ay=0 \rightarrow y=\frac{1}{a}$$

$$\rightarrow y=\frac{1}{a}$$

$$\therefore \frac{1}{y \cdot 1} + \frac{1}{(1-ay) \frac{1}{a}}$$

Solution method: $ax+by+c$ এটাকে v ধরব, উভয়
পক্ষকে x এর সাপেক্ষে অন্তরীকরণ করব, ওখানে

যেহে $\frac{dy}{dx}$ এর মান হবে v ধরব।

এবার $\frac{dy}{dx}$ এর মান ও যেটা সমান v ধরছিলাম তা
given eqn এ বসাই দিবা,

তারপর variable separable করব, তারপর integration
করব,

working rule: $\frac{dy}{dx} = f(ax+by+c) \text{ ————— (1)}$

Let, $ax+by+c = v$

so, $a+b \frac{dy}{dx} = \frac{dv}{dx}$

or, $\frac{dy}{dx} = \frac{1}{b} \left(\frac{dv}{dx} - a \right)$

From eqn (1), $\frac{1}{b} \left(\frac{dv}{dx} - a \right) = f(v)$

or, $\frac{dv}{dx} = bf(v) + a$

or, $\frac{dv}{bf(v)+a} = dx$

Now integrating we will get our required
solution.

Ex-1) solve, $\frac{dy}{dx} = (4x+y+1)^2$

Solution: Given that,

$$\frac{dy}{dx} = (4x+y+1)^2 \quad \text{--- (1)}$$

Let, $4x+y+1=v$

$$\text{or, } 4 + \frac{dy}{dx} = \frac{dv}{dx}$$

$$\text{or, } \frac{dy}{dx} = \frac{dv}{dx} - 4$$

so eqn (1) become,

$$\frac{dv}{dx} - 4 = v^2$$

$$\text{or, } \frac{dv}{dx} = v^2 + 4$$

$$\text{or, } \frac{dv}{4+v^2} = dx$$

$$\text{or, } \frac{dv}{2^2+v^2} = dx$$

Now Integrating we get, $\frac{1}{2} \tan^{-1} \frac{v}{2} = x + C$

$\therefore \frac{1}{2} \tan^{-1} \left(\frac{4x+y+1}{2} \right) = x + C$; which is the required solution.

Ex-2 | solve $\frac{dy}{dx} = \sin(x+y) + \cos(x+y)$

Soln : Given that,

$$\frac{dy}{dx} = \sin(x+y) + \cos(x+y) \quad \text{--- (1)}$$

Let, $x+y = v$

$$\text{or, } 1 + \frac{dy}{dx} = \frac{dv}{dx}$$

$$\text{or, } \frac{dy}{dx} = \frac{dv}{dx} - 1$$

So eqn (1) become, $\frac{dv}{dx} - 1 = \sin v + \cos v$

$$\text{or, } \frac{dv}{\sin v + \cos v + 1} = dx$$

$$\text{or, } \frac{dv}{1 + \cos v + \sin v} = dx$$

$$\text{or, } \frac{dv}{2\cos^2 \frac{v}{2} + 2\sin \frac{v}{2} \cos \frac{v}{2}} = dx$$

$$\text{or, } \frac{dv}{2\cos^2 \frac{v}{2} (1 + \tan \frac{v}{2})} = dx$$

$$\text{or, } \left(\frac{\frac{1}{2} \sec^2 \frac{v}{2}}{1 + \tan \frac{v}{2}} \right) dv = dx$$

$$\int \frac{f'(x)}{f(x)} dx = \ln(f(x))$$

$$\begin{aligned} 1 + \cos 2A &= 2\cos^2 A \\ \sin 2A &= 2\sin A \cos A \end{aligned}$$

Now Integrating ;

$$\ln \left(1 + \tan \frac{y}{2} \right) = x + C$$

$$\therefore \ln \left(1 + \tan \left(\frac{x+y}{2} \right) \right) = x + C$$

which is the required solution.

Ex-3 | Solve $(x-y)^2 \frac{dy}{dx} = a^2$

Soln: Given that,

$$(x-y)^2 \frac{dy}{dx} = a^2 \quad \text{--- (1)}$$

Let, $x-y = v$

$$\text{or, } 1 - \frac{dy}{dx} = \frac{dv}{dx}$$

$$\text{or, } \frac{dy}{dx} = 1 - \frac{dv}{dx}$$

So eqn (1) become, $v^2 \left(1 - \frac{dv}{dx} \right) = a^2$

$$\text{or, } \frac{dv}{dx} = 1 - \frac{a^2}{v^2}$$

$$\text{or, } \left(\frac{v^2 - a^2}{v^2} \right) dv = \text{or, } \left(\frac{v^2}{v^2 - a^2} \right) dv = dx$$

$$\text{or, } \left(1 + \frac{a^2}{v^2 - a^2} \right) dv = dx$$

Now Integrating,

$$r+a^2 \cdot \frac{1}{2a} \ln\left(\frac{r-a}{r+a}\right) = x + C$$

$$\text{or, } x+C = (x-y) + \frac{a}{2} \ln \frac{x-y-a}{x-y+a}$$

which is the required solution.

Ex-4] solve $(x+y)^2 \frac{dy}{dx} = a^2$

Solⁿ: Given,

$$(x+y)^2 \frac{dy}{dx} = a^2 \quad \text{--- (1)}$$

Let, $(x+y) = v$

$$\text{or, } 1 + \frac{dy}{dx} = \frac{dv}{dx}$$

$$\text{or, } \frac{dy}{dx} = \frac{dv}{dx} - 1$$

So eqn (1) become, $v^2 \left(\frac{dv}{dx} - 1 \right) = a^2$

$$\text{or, } \frac{dv}{dx} = 1 + \frac{a^2}{v^2}$$

$$\text{or, } \left(\frac{v^2}{v^2+a^2} \right) dv = dx$$

$$\text{or, } \left(\frac{v^2+a^2-a^2}{v^2+a^2} \right) dv = dx$$

$$\left(1 - \frac{a}{v^2 + av}\right) dv = dx$$

Now Integrating,

$$v - a^2 \cdot \frac{1}{a} \tan^{-1} \frac{v}{a} = x + C$$

$$\text{or, } v - a \tan^{-1} \frac{v}{a} = x + C$$

$$\text{or, } x + y - a \tan^{-1} \frac{x+y}{a} = x + C$$

or, $y = C + a \tan^{-1} \frac{x+y}{a}$; which is the required solution.

Ex: 6 | solve $x \frac{dy}{dx} - y = x \sqrt{x^2 + y^2}$

Soln: Given,

$$x \frac{dy}{dx} - y = x \sqrt{x^2 + y^2}$$

Ex: 7 | solve $\left(\frac{x+y-a}{x+y-b} \right) \frac{dy}{dx} = \frac{x+y+a}{x+y+b}$

Soln: Given,

$$\frac{x+y-a}{x+y-b} \Rightarrow \left(\frac{x+y-a}{x+y-b} \right) \frac{dy}{dx} = \frac{x+y+a}{x+y+b} \quad (1)$$

Let, $x+y = v$

$$\text{or, } 1 + \frac{dy}{dx} = \frac{dv}{dx}$$

$$\text{or, } \frac{dy}{dx} = \frac{dv}{dx} - 1$$

so eqn (1) become,

$$\left(\frac{v-a}{v-b} \right) \left(\frac{dv}{dx} - 1 \right) = \frac{v+a}{v+b}$$

$$\begin{aligned} \text{or, } \frac{dv}{dx} &= 1 + \left(\frac{v+a}{v+b} \right) \left(\frac{v-b}{v-a} \right) = \frac{v^2 + vb - va - ab + v^2 - vb + va - ab}{v^2 + v(b-a) - ab} \\ &= \frac{2(v^2 - ab)}{v^2 + v(b-a) - ab} \end{aligned}$$

$$\text{or, } \left\{ \frac{v^2 + v(b-a) - ab}{2(v^2 - ab)} \right\} dv = dx$$

$$\text{or, } \left\{ \frac{(v^2 - ab) + v(b-a)}{v^2 - ab} \right\} dv = 2dx$$

$$\text{or, } \left\{ 1 + \frac{b-a}{2} \frac{2v}{v^2 - ab} \right\} dv = 2dx$$

Now Integrating, $v + \frac{b-a}{2} \ln(v^2 - ab) = 2x + c$

$$\therefore x + y + \frac{b-a}{2} \ln \left[(x+y)^2 - ab \right] = 2x + c$$

which is the required solution.

Homogeneous Differential Equation:

Homogeneous: যে eqn এর প্রত্যেক পদের চলক
দ্বারা সমান, সমান.

working rule: If any equation of the
form $\frac{dy}{dx} = \frac{f_1(x,y)}{f_2(x,y)}$ in which $f_1(x,y)$ &

$f_2(x,y)$ are homogeneous function
of x & y then putting

$$y = vx$$

এটাকে x এর সাপেক্ষে অন্তরীকরণ করে $\frac{dy}{dx}$
এর মান বের করা হবে.

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

Given eqn এ মানগুলো বসিয়ে দিবে

তখন variable separable করে Integration
করা হবে.

Ex-7 $(x \cos \frac{y}{x} + y \sin \frac{y}{x}) y = (y \sin \frac{y}{x} - x \cos \frac{y}{x})$
 $x \frac{dy}{dx}$

Soln: Given that,

$$(x \cos \frac{y}{x} + y \sin \frac{y}{x}) y = (y \sin \frac{y}{x} - x \cos \frac{y}{x})$$
$$x \frac{dy}{dx}$$

$$\text{or, } \frac{dy}{dx} = \frac{y \left(x \cos \frac{y}{x} + y \sin \frac{y}{x} \right)}{x \left(y \sin \frac{y}{x} - x \cos \frac{y}{x} \right)} \quad (1)$$

Let, $y = vx$

$$\text{or, } \frac{dy}{dx} = v + x \frac{dv}{dx}$$

Now, eqn(1) become,

$$v + x \frac{dv}{dx} = \frac{vx \left(x \cos \frac{vx}{x} + vx \sin \frac{vx}{x} \right)}{x \left(vx \sin \frac{vx}{x} - x \cos \frac{vx}{x} \right)}$$

$$\text{or, } x \frac{dv}{dx} = \frac{v (\cos v + v \sin v)}{v \sin v - \cos v} - v$$

$$= \frac{v (\cos v + v \sin v) - v (v \sin v - \cos v)}{v \sin v - \cos v}$$

$$= \frac{2v \cos v}{v \sin v - \cos v}$$

$$\text{or, } \left(\frac{v \sin v - \cos v}{v \cos v} \right) dv = \frac{2}{x} dx$$

$$\text{or, } \left(\tan v - \frac{1}{v} \right) dv = \frac{2}{x} dx$$

$$\int \tan x dx = \ln |\sec x| + C$$

Now Integrating,

$$\ln \sec v - \ln v = 2 \ln x + \ln C$$

$$\text{or, } \ln \frac{\sec v}{v} = \ln c x^2$$

$$\text{or, } c x^2 = \frac{\sec v}{v}$$

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$$\text{or, } c x^2 = \frac{\sec y/x}{y/x}$$

or, $cxy = \sec y/x$; which is required solution.

Ex-1 | Solve $(x^2 + y^2)dx + 2xy dy = 0$

Soln : Given that,

$$(x^2 + y^2)dx + 2xy dy = 0$$

$$\text{or, } \frac{dy}{dx} = - \frac{x^2 + y^2}{2xy} \quad \text{--- (1)}$$

Let, $y = vx$

$$\text{or, } \frac{dy}{dx} = v + x \frac{dv}{dx}$$

Now eqn (1) become,

$$v + x \frac{dv}{dx} = - \frac{x^2 + v^2 x^2}{2x \cdot vx} = - \frac{1 + v^2}{2v}$$

$$x \frac{dv}{dx} = - \frac{1 + v^2}{2v} - v = \frac{-1 - v^2 - 2v^2}{2v}$$

$$\text{or, } x \frac{dv}{dx} = - \frac{1+3v^2}{2v}$$

$$\text{or, } \frac{dx}{x} = - \frac{2v}{1+3v^2} dv$$

$$= - \frac{1}{3} \cdot \frac{6v}{1+3v^2} dv$$

$$\text{Now Integrating, } \ln x = - \frac{1}{3} \ln(1+3v^2) + \ln c$$

$$\text{or, } \ln x + \frac{1}{3} \ln(1+3v^2) = \ln c$$

$$\text{or, } \ln x + \ln(1+3v^2)^{1/3} = \ln c$$

$$\text{or, } x(1+3v^2)^{1/3} = c$$

$$\text{or, } x \left(1 + 3 \frac{y^2}{x^2}\right)^{1/3} = c$$

which is required solution.

Ex-2 | Solve $x^2y dx - (x^3 + y^3) dy = 0$

Soln: Given,

$$x^2y dx - (x^3 + y^3) dy = 0$$

$$\text{or, } (x^3 + y^3) dy = x^2y dx$$

$$\text{or, } \frac{dy}{dx} = \frac{x^2y}{x^3 + y^3} \quad \text{--- (1)}$$

Let, $y = vx$

$$\therefore \frac{dy}{dx} = v + x \frac{dv}{dx}$$

Now eqn (1) become,

$$v + x \frac{dv}{dx} = \frac{x^2 \cdot vx}{x^3 + v^3 x^3} = \frac{v}{1+v^3}$$

$$\text{or, } x \frac{dv}{dx} = \frac{v}{1+v^3} - v = \frac{v - v - v^4}{1+v^3} = -\frac{v^4}{1+v^3}$$

$$\text{or, } \frac{dx}{x} = \left(-\frac{1+v^3}{v^4} \right) dv$$

$$\therefore \frac{dx}{x} = \left[-\left(\frac{1}{v^4} + \frac{1}{v} \right) \right] dv$$

$$\begin{aligned} \text{Now Integrating, } \ln x &= -\left(-\frac{1}{3v^3} + \ln v \right) + C \\ &= \frac{1}{3v^3} - \ln v + C \end{aligned}$$

$$\text{or, } \ln x = \frac{x^3}{3y^3} - \ln \frac{y}{x} + C$$

$$\text{or, } 0 = \frac{x^3}{3y^3} - \ln y + C$$

$$\therefore \ln y = \frac{x^3}{3y^3} + C$$

which is required solution.

Ex-3 solve $\frac{dy}{dx} = \frac{y^3 + 3x^2y}{x^3 + 3xy^2}$

Soln: Given that, $\frac{dy}{dx} = \frac{y^3 + 3x^2y}{x^3 + 3xy^2}$ (1)

Let, $y = vx$

$\therefore \frac{dy}{dx} = v + x \frac{dv}{dx}$

Now eqn (1) become, $v + x \frac{dv}{dx} = \frac{v^3x^3 + 3x^2 \cdot vx}{x^3 + 3x \cdot v^2x^2}$

$= \frac{v^3 + 3v}{1 + 3v^2}$

or, $x \frac{dv}{dx} = \frac{v^3 + 3v}{1 + 3v^2} - v = \frac{v^3 + 3v - v - 3v^3}{1 + 3v^2}$

$= \frac{2(v - v^3) \cdot 2v(1 - v^2)}{1 + 3v^2}$

or, $\frac{2}{x} dx = \frac{1 + 3v^2}{2v(1 - v^2)} dv$

or, $\frac{2}{x} dx = \left(\frac{1}{v} - \frac{2}{1+v} + \frac{2}{1-v} \right) dv$

Now integrating,

$2 \ln x + \cancel{+ \ln c} = \ln v - 2 \ln(1+v) + 2 \ln(1-v) + \ln c$

$\ln x^2 + \cancel{+ \ln c} = \ln v - \ln(1+v)^2 - \ln(1-v)^2 + \ln c$

$= \ln \frac{cv}{(1+v)^2(1-v)^2}$

$v=0 \therefore \frac{1+3v^2}{(1+v)(1-v)} = 1$

$1+v=0 \therefore v=-1$

$\frac{1+3v^2}{v(1-v)} = -2$

$1-v=0 \therefore v=1$

$\frac{1+3v^2}{v(1+v)} = 2$

$$\text{or, } x^2(1+v)^2(1-v)^2 = c v$$

$$\text{or, } x^2\left(1+\frac{y}{x}\right)^2\left(1-\frac{y}{x}\right)^2 = c \frac{y}{x}$$

$$\therefore x^3\left(1+\frac{y}{x}\right)^2\left(1-\frac{y}{x}\right)^2 = y c$$

which is required solution.

Ex-4] solve, $y^2 + x^2 \frac{dy}{dx} = xy \frac{dy}{dx}$

Soln: Given that,

$$y^2 + x^2 \frac{dy}{dx} = xy \frac{dy}{dx}$$

$$\text{or, } \frac{dy}{dx} = \frac{y^2}{x^2 xy - x^2} \quad (1)$$

Let, $y = vx \therefore \frac{dy}{dx} = v + x \frac{dv}{dx}$

Now eqn (1) become,

$$v + x \frac{dv}{dx} = \frac{v^2 x^2}{x \cdot vx - x^2} = \frac{v^2}{v-1}$$

$$\text{or, } x \frac{dv}{dx} = \frac{v^2}{v-1} - v = \frac{v^2 - v^2 + v}{v-1} = \frac{v}{v-1}$$

$$\text{or, } \frac{dx}{x} = \left(\frac{v-1}{v}\right) dv = \left(1 - \frac{1}{v}\right) dv$$

Now Integrating, $\ln x = (v - \ln v) + \ln c$